

CASE STUDY · CONVERSATIONAL AI · AWS

RestaurantX Conversational AI Ordering System

Amazon Lex + Lambda + Comprehend + Polly + S3

A table-side web app and mobile ordering platform powered by a sentiment-aware AI chatbot. Customers order by voice or text, pay by card, and receive audio confirmation all without waiter involvement.

Amazon Lex

AWS Lambda

Comprehend

Polly

S3

Voice Orders

Payments

AWS

OVERVIEW

Project Summary

RestaurantX had a problem every busy restaurant knows well. During lunch and dinner rushes, waitstaff were stretched thin taking orders manually, mistakes crept in, and customers at full tables sometimes waited too long before anyone came to them. There was also no way for regular customers to order in advance from outside the restaurant.

The solution was a fully integrated AI ordering system built on AWS. Every table in the restaurant got a tablet running a web app connected to an Amazon Lex chatbot. Customers browse the full menu, place their order by typing or speaking, confirm by card, and receive a text and audio summary of what they ordered. The kitchen gets the order automatically once payment clears. A mobile app extends the same experience to customers ordering from home.

5 AWS Services	2 Channels (Web + Mobile)	Voice Ordering Live	Card Payments Integrated
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ARCHITECTURE

System Design

The system is built entirely on AWS managed services. There are no servers to maintain and the infrastructure scales automatically during peak hours. The architecture separates concerns cleanly: Amazon Lex owns the conversation, Lambda owns the logic, Comprehend handles sentiment in real time, Polly produces audio, and S3 stores and serves the files.

1 Customer (Web / Mobile) Tablet tablet or mobile app	2 Amazon Lex Intent + slot management	3 AWS Lambda Backend orchestration	4 Comprehend + Polly Sentiment + voice TTS	5 Payment Gateway Card processing	6 Amazon S3 Audio file storage	7 Order to Kitchen Confirmed + paid order
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Lambda and Comprehend (highlighted) form the AI core. Every customer message passes through Lambda, which calls Comprehend for sentiment before deciding how Lex should respond.

WORKFLOW

Step by Step

Step 1	Customer Opens the App On the table tablet or mobile app, the customer opens the chat interface. The chatbot greets them, presents the menu, and begins collecting order details through a guided conversation. The customer can type or speak.
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Step 2	Slot Filling via Amazon Lex Lex manages the conversation flow. It identifies the customer's intent (placing an order) and collects all required details: pizza type, size, crust, toppings, quantity, special instructions, and contact number for mobile orders. If anything is missing or ambiguous, Lex prompts the customer again.
Step 3	Real Time Sentiment Analysis As the customer types or speaks, each message is sent to Amazon Comprehend via Lambda. Comprehend returns a sentiment label (Positive, Neutral, or Negative). Lambda uses this to shape the tone of the chatbot's next reply, making the conversation feel human and responsive to the customer's mood.
Step 4	Order Confirmation Once all slots are filled, the chatbot presents a complete order summary and asks the customer to confirm. The customer can review and make changes at this stage before committing.
Step 5	Card Payment After confirmation, the customer is prompted to pay by credit or debit card. Lambda handles the payment flow. The order is only dispatched to the kitchen after the charge is successful, ensuring no unpaid orders reach the kitchen.
Step 6	Voice Confirmation via Polly and S3 Lambda calls Amazon Polly to generate a natural-sounding audio summary of the confirmed order. The audio file is uploaded to Amazon S3 and a secure link is returned to the customer. They receive both a text confirmation and a playable audio summary.
Step 7	Order Sent to Kitchen The confirmed, paid order is pushed to the kitchen management system. Staff see the full order details, the table number or customer contact, and any special instructions — all without a waiter having taken the order manually.

KEY FEATURES

What the System Does

Table Web App	Tablet-based ordering at every table. No waiter needed for standard orders. Direct to kitchen once payment confirms.
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Mobile App Ordering	Same chatbot and menu experience for customers ordering from home or on the go. Available on both iOS and Android.
Voice and Text Input	Customers can type or speak their order. Voice ordering is fully implemented and live across both channels.
Sentiment Aware Tone	Amazon Comprehend analyses every message in real time. Positive input gets enthusiasm, negative input gets empathy. The chatbot adjusts automatically.
Card Payment	Credit and debit card payment collected as part of the chatbot conversation. Orders only reach the kitchen after successful payment.
Audio Confirmation	Amazon Polly converts the order summary to speech. Stored in S3 and delivered as a playable link alongside the text confirmation.

SENTIMENT RESPONSE MODEL

How the Chatbot Reads the Room

Every customer message is classified by Amazon Comprehend before a response is generated. The chatbot does not give the same reply to someone excited about their order and someone frustrated by a delay. The tone is chosen based on what the customer is actually feeling.

Sentiment	Example Customer Message	Chatbot Response Style
Positive	This looks amazing, I will take the Margherita!	Enthusiastic and warm. Great choice, coming right up.
Neutral	I want a large pepperoni.	Clear and efficient. Got it, large pepperoni noted.
Negative	This is taking too long.	Empathetic and helpful. I am sorry for the wait. Let me sort this out right away.

AWS SERVICES

Technology Stack

Amazon Lex	Conversational interface layer. Handles intent recognition, dialog management, and slot elicitation across both voice and text input.
AWS Lambda	Backend orchestration. Receives Lex events, calls Comprehend and Polly, processes payments, and sends confirmed orders to the kitchen.
Amazon Comprehend	Real time sentiment analysis on every customer message. Returns Positive, Neutral, or Negative, which Lambda uses to shape the chatbot's tone.

Amazon Polly	Text to speech engine. Converts the order confirmation text into a natural audio file immediately after the order is placed.
Amazon S3	Stores Polly audio files and serves them through secure links. Customers receive the link in their confirmation message for instant playback.

INTERVIEW TALKING POINTS

Key Topics to Present

→ Serverless Architecture on AWS

The entire system runs without managing a single server. Lambda scales automatically during rush hours, Lex handles concurrent conversations across every table simultaneously, and the infrastructure cost is proportional to actual usage.

→ Real Time Sentiment Analysis in a Live Product

Sentiment analysis is not a reporting feature here. It runs on every message in the active conversation and changes what the customer sees next. The Comprehend call adds negligible latency while meaningfully improving the experience.

→ Multi-Channel Conversational AI

The same Lex bot, Lambda function, and backend logic serves both the table web app and the mobile app. Building once for multiple channels is a direct cost and maintenance advantage.

→ Closing the Loop with Voice Confirmation

Polly is not just a novelty feature. For customers who placed a voice order, hearing a voice confirmation closes the interaction in a natural and accessible way. The S3 link also works as a receipt the customer can replay later.

→ Payment as Part of the Conversation

Integrating card payment directly into the chatbot flow removes a handoff point. The customer does not switch to a separate payment screen. The order and payment happen in one continuous conversation, which reduces abandonment.

→ Business Impact

The system removes the waiter from the standard order flow entirely. Table turn time improves, order errors drop, and kitchen staff see confirmed, paid orders only. The mobile app extends the restaurant's reach beyond the physical premises.

FUTURE ROADMAP

Scheduled push notifications for lunch (12 pm) and dinner (7 pm) windows are the next planned feature, prompting mobile customers to order within their meal window. Additional roadmap items include full hands-free voice ordering, returning customer reorder shortcuts, AI-driven menu recommendations based on past orders, and multilingual support.